

Enterprise Workforce Management Platform with AI Operations Assistant

1. Project Overview

1.1 Introduction

Organizations today manage hundreds or even thousands of employees across multiple departments, projects, and office locations. As businesses grow, managing employee information manually using spreadsheets, emails, and paper documents becomes inefficient and error-prone.

The **Enterprise Workforce Management Platform** is designed to digitize and automate the complete employee lifecycle—from recruitment to retirement—through a centralized web application.

The system enables organizations to manage employees, departments, attendance, leave requests, payroll, performance reviews, projects, assets, and support tickets from a single platform.

To further enhance productivity, the application integrates an **AI Operations Assistant** capable of answering HR-related questions, summarizing documents, explaining company policies, assisting in recruitment, and providing workforce insights.

The platform should be modular, scalable, secure, and user-friendly while following enterprise software development practices.

2. Business Problem

Most small and medium-sized organizations use multiple disconnected systems to manage HR operations.

Current challenges include:

- Employee records stored in spreadsheets
- Attendance tracked manually
- Leave requests handled via email
- Payroll calculated using separate software
- Documents stored in folders
- Recruitment tracked through emails
- No centralized analytics
- Slow approval processes
- Lack of employee self-service
- Heavy dependency on HR teams

These problems result in increased operational costs, delayed decision-making, reduced productivity, and poor employee experience.

3. Proposed Solution

Develop a centralized Enterprise Workforce Management Platform that provides:

- Employee Management
 - Recruitment Management
 - Attendance Tracking
 - Leave Management
 - Payroll Management
 - Project & Task Management
 - Asset Tracking
 - Document Management
 - Help Desk System
 - AI Operations Assistant
 - Analytics Dashboard
 - Real-Time Notifications
 - Role-Based Access Control
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4. Project Objectives

Objective ID	Objective
OBJ-01	Automate employee management
OBJ-02	Digitize attendance and leave process
OBJ-03	Reduce payroll processing time
OBJ-04	Improve recruitment workflow
OBJ-05	Provide centralized document storage
OBJ-06	Introduce AI-powered HR assistance
OBJ-07	Improve organizational productivity
OBJ-08	Generate analytical reports
OBJ-09	Ensure secure role-based access
OBJ-10	Build an enterprise-ready scalable application

5. Expected Benefits

Benefit	Description
Reduced Manual Work	Eliminate paperwork and spreadsheets
Improved Accuracy	Automated calculations reduce human error
Faster Approvals	Automated workflows speed up approvals
Better Communication	Notifications keep users informed

Benefit	Description
Centralized Data	Single source of truth for employee information
AI Assistance	Faster answers to HR and policy questions
Better Decision Making	Analytics dashboards provide insights
Improved Security	Role-based access protects sensitive information

6. Project Scope

In Scope

The project will include the following modules:

Module	Included
Authentication	✓
Organization Management	✓
Employee Management	✓
Recruitment	✓
Attendance	✓
Leave Management	✓
Payroll	✓
Performance Management	✓
Project Management	✓
Asset Management	✓
Help Desk	✓
Document Management	✓
AI Assistant	✓
Reports & Analytics	✓
Notifications	✓

Out of Scope

The following features are excluded from this version:

- Mobile Application
- Biometric Device Integration
- Accounting Software Integration
- Government Tax Filing
- Banking APIs
- Video Conferencing
- Multi-language Support
- Offline Functionality

7. Stakeholders

Stakeholder	Responsibility
Super Admin	Overall system administration
Organization Admin	Manage organization configuration
HR Manager	Employee and recruitment management
Department Manager	Manage department and approvals
Team Lead	Assign tasks and review employees
Employee	Daily operations and self-service
Finance Team	Payroll management
IT Administrator	Technical support
Auditor	Compliance and audit review

8. User Roles

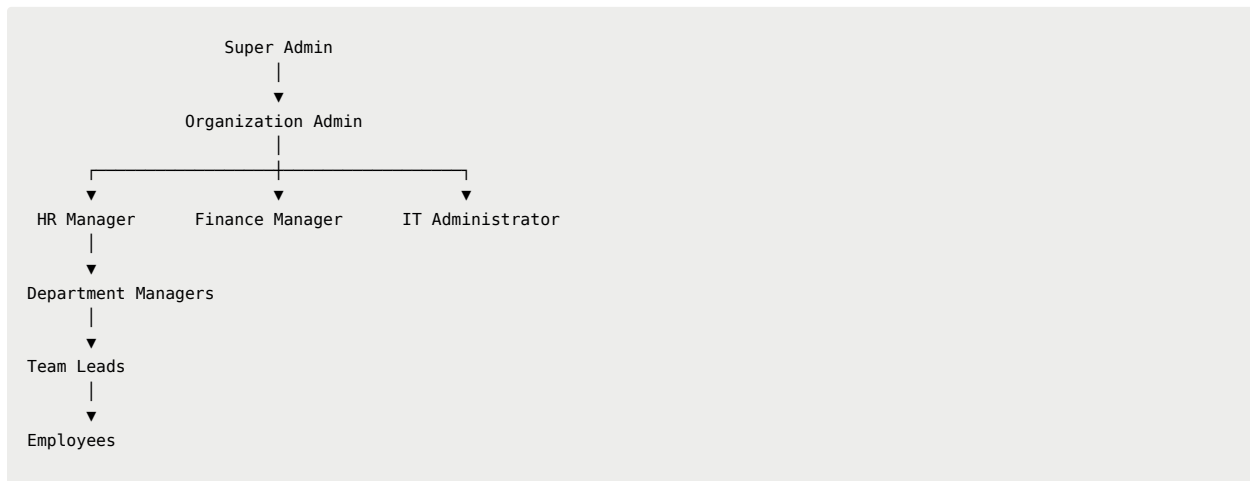
Role	Primary Responsibility
Super Admin	Complete control over the platform
Organization Admin	Configure organization settings
HR Manager	Employee lifecycle management
Manager	Team management and approvals
Team Lead	Project and task management
Employee	Self-service portal
Finance Executive	Payroll operations
IT Administrator	Help Desk and asset tracking
Auditor	Read-only access for auditing

9. Role-Based Access Control (RBAC)

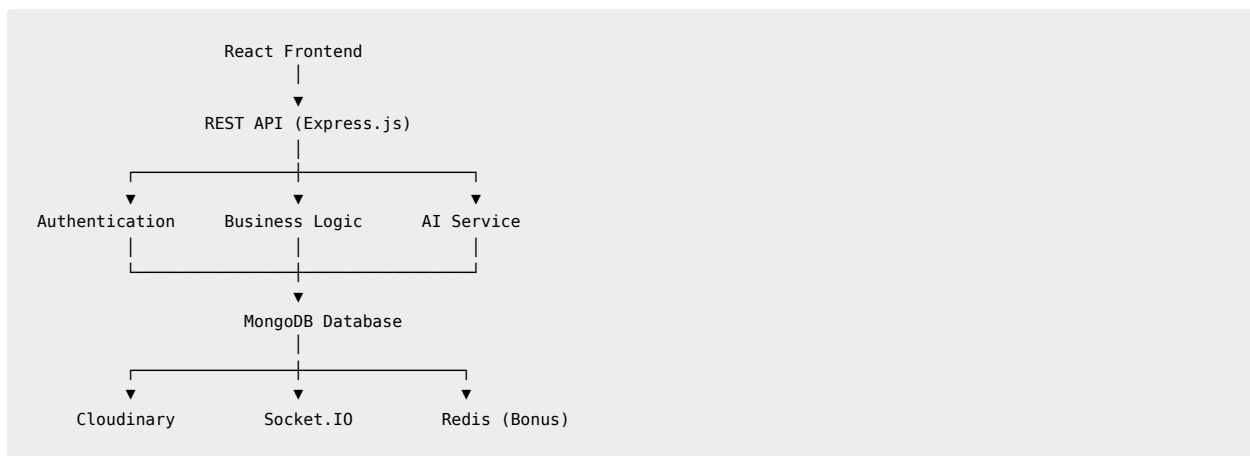
Module	Super Admin	HR	Manager	Employee	Finance	IT
User Management	Full	Read	No	No	No	No
Employee Management	Full	CRUD	Read	Self	Read	Read
Recruitment	Full	CRUD	Interview	No	No	No
Attendance	Full	CRUD	Approve	Mark Attendance	Read	No
Leave	Full	CRUD	Approve	Apply	Read	No
Payroll	Full	Read	Read	View Own	CRUD	No
Project Management	Full	Read	CRUD	Assigned Tasks	No	No

Module	Super Admin	HR	Manager	Employee	Finance	IT
Asset Management	Full	Read	Read	View Assigned	No	CRUD
Help Desk	Full	Read	Read	Raise Ticket	No	CRUD
Reports	Full	Generate	Department	Personal	Payroll	IT Reports

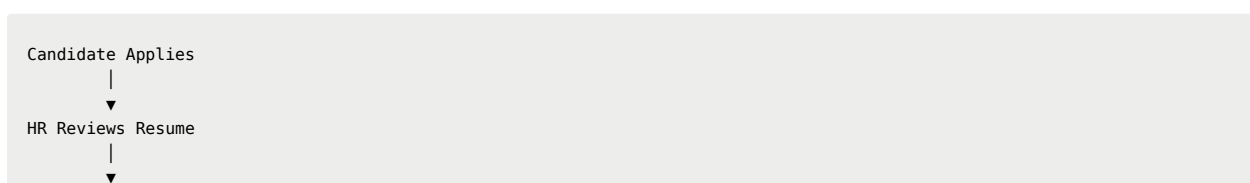
10. Organization Structure

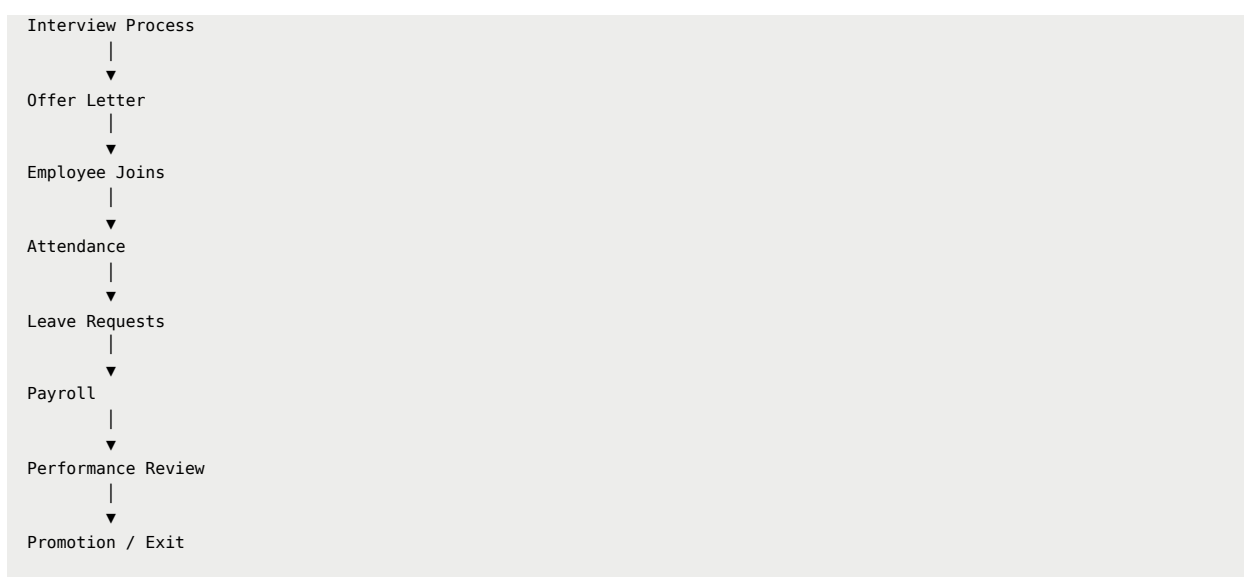


11. High-Level System Architecture



12. Core Business Workflow





13. Key Functional Modules

Module ID	Module Name	Description
M-01	Authentication	User registration, login, security
M-02	Organization Management	Departments, designations, hierarchy
M-03	Employee Management	Employee lifecycle management
M-04	Recruitment	Candidate management
M-05	Attendance	Daily attendance tracking
M-06	Leave	Leave application and approval
M-07	Payroll	Salary processing
M-08	Performance	Employee reviews and KPIs
M-09	Project Management	Projects, tasks, assignments
M-10	Asset Management	Company assets
M-11	Help Desk	Ticket management
M-12	Document Management	Secure document repository
M-13	Notifications	Email, in-app, real-time alerts
M-14	Reports & Analytics	Business dashboards
M-15	AI Operations Assistant	AI-powered HR assistant

14. AI Operations Assistant

The AI Assistant should be accessible from every page of the application and provide contextual assistance based on the user's role.

AI Features

Feature	Description
HR Policy Assistant	Answers questions about company policies
Resume Analyzer	Summarizes resumes and highlights skills
Attendance Insights	Explains attendance patterns
Leave Assistant	Guides employees on leave policies
Payroll Explainer	Breaks down salary calculations
Document Search	Retrieves information from uploaded documents
Employee Performance Summary	Summarizes KPIs and reviews
Meeting Notes Summarizer	Converts meeting notes into concise summaries

Example Interaction

Employee Prompt:

How many casual leaves do I have remaining?

AI Response:

You have 5 casual leaves available.
You have used 7 out of your allocated 12 casual leaves this year.

15. Assumptions

- Users will access the application through a modern web browser.
- HR creates employee accounts before first login.
- Cloud storage is available for document uploads.
- AI services are accessed through external APIs.
- Email notifications are configured and functional.
- Each employee belongs to exactly one department and reports to one manager.

16. Constraints

Constraint	Description
Technology	MERN Stack only
Database	MongoDB
Hosting	Free-tier or student-friendly cloud hosting
Timeline	Must fit within an academic semester
AI	External AI API usage may be rate-limited

17. Success Criteria

The project will be considered successful if:

- All core modules are implemented and integrated.
 - Role-based access control is enforced correctly.
 - Employees can complete common HR tasks through self-service.
 - HR can manage recruitment, attendance, payroll, and employee records efficiently.
 - AI Assistant provides meaningful responses based on organizational data.
 - Dashboards and reports provide actionable insights.
 - The application is deployed online with complete source code and documentation.
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Part 2 – Functional Requirements (Core Modules)

Module 1 – Authentication & User Management

1.1 Module Overview

Authentication is the entry point of the application. It ensures that only authorized users can access the system while protecting organizational data through secure authentication and role-based authorization.

The module should support different user roles and ensure that each user only has access to the features assigned to their role.

Business Objective

Develop a secure authentication system that protects organizational data while providing a seamless login experience for all users.

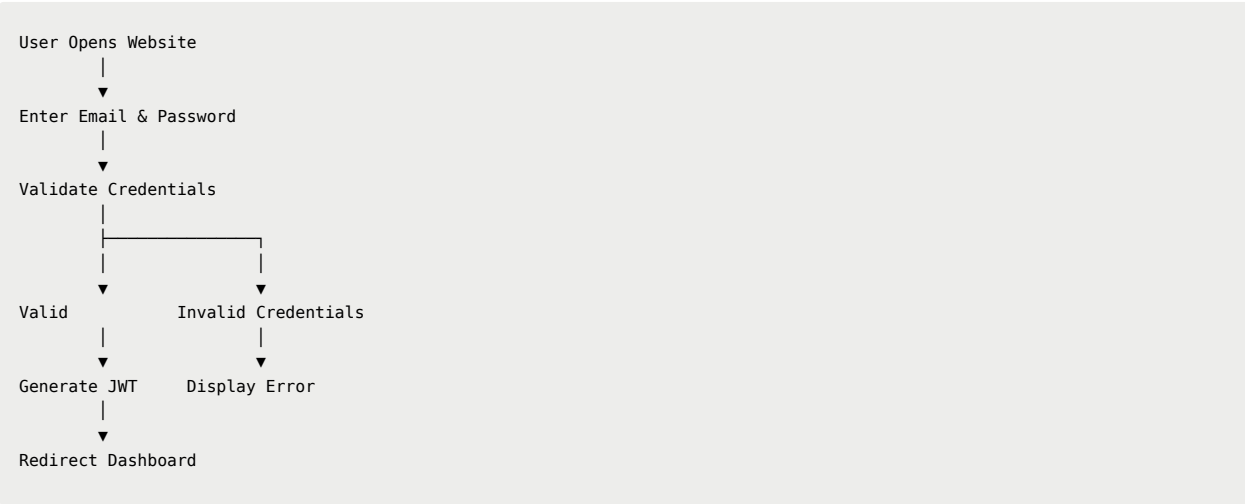
Actors

Actor	Responsibilities
Super Admin	Create organizations and administrators
Organization Admin	Manage users within the organization
HR Manager	Create employee accounts
Employee	Login and manage own profile
IT Administrator	Reset passwords and unlock accounts

Functional Requirements

ID	Requirement
FR-A01	User Login
FR-A02	Forgot Password
FR-A03	Reset Password
FR-A04	Change Password
FR-A05	JWT Authentication
FR-A06	Refresh Token
FR-A07	Role Based Authorization
FR-A08	Session Timeout
FR-A09	Profile Management
FR-A10	Account Lock after multiple failed attempts

Login Workflow



Business Rules

Rule ID	Description
BR-01	Email address must be unique
BR-02	Password minimum length 8 characters
BR-03	Password must contain uppercase, lowercase, number and special character
BR-04	Account locks after 5 failed login attempts
BR-05	Only HR/Admin can create employee accounts
BR-06	JWT expires after configurable duration
BR-07	Refresh Token required for new access token

Validation Rules

Field	Validation
Email	Valid Email Format
Password	Required
Confirm Password	Must Match
Mobile Number	10 digits
Name	Required

Sample Login JSON

```
{
  "email": "john.doe@company.com",
  "password": "Secure@123"
}
```

Successful Response

```
{
  "success": true,
  "token": "JWT_TOKEN",
  "role": "HR_MANAGER",
  "userId": "EMP001"
}
```

Failed Response

```
{
  "success": false,
  "message": "Invalid Credentials"
}
```

Notifications

Event	Recipient
New Login	User
Password Changed	User
Password Reset	User
Suspicious Login	Admin

Edge Cases

Scenario	Expected Behaviour
Invalid Password	Reject Login
Locked Account	Display Contact Admin
Expired Token	Force Login
Deleted User	Access Denied

Acceptance Criteria

- ✓ User logs in successfully
- ✓ Invalid credentials rejected
- ✓ JWT generated
- ✓ Role permissions loaded
- ✓ Login history stored

Module 2 – Organization Management

Module Overview

This module manages the organizational structure including departments, designations, office locations, work shifts and reporting hierarchy.

The organization should be able to configure itself without changing application code.

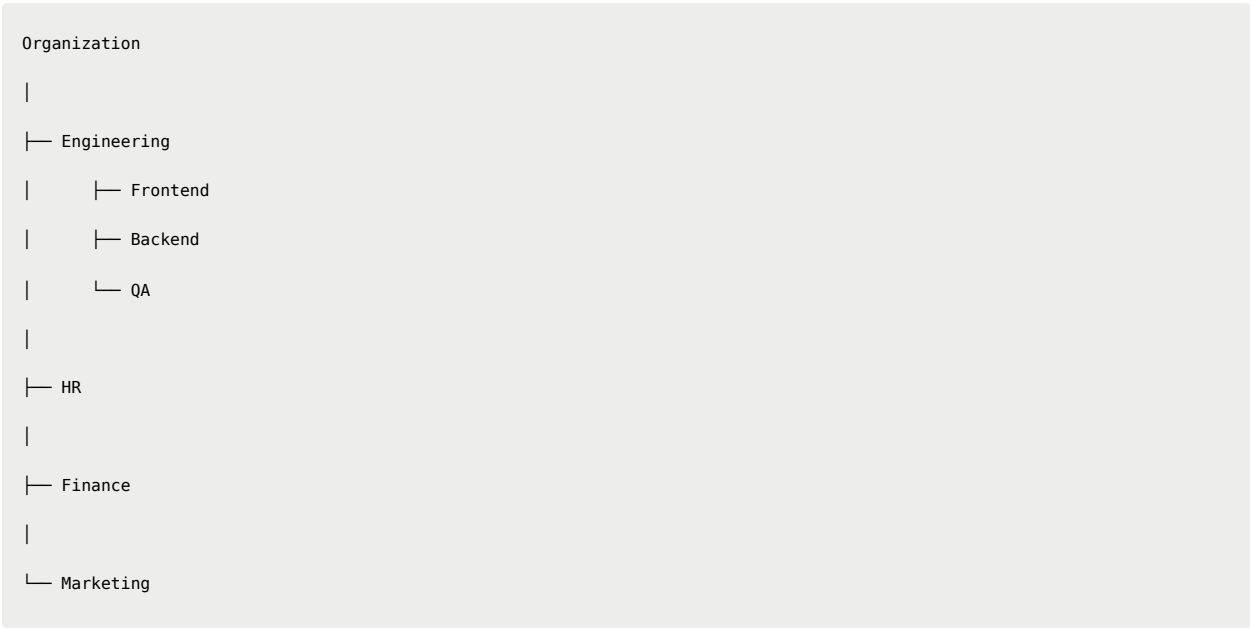
Business Objective

Provide a flexible structure capable of representing organizations of different sizes.

Features

Feature	Description
Create Organization	
Manage Departments	
Create Designations	
Office Locations	
Holiday Calendar	
Work Shifts	
Reporting Hierarchy	

Department Hierarchy



Department JSON

```
{
  "departmentName": "Engineering",
  "departmentCode": "ENG",
  "manager": "EMP0045",
  "employees": 82,
  "status": "Active"
}
```

Business Rules

Rule	Description
Department Code Unique	
One Department Manager	
Department cannot be deleted if employees exist	
Department Name Required	

Acceptance Criteria

- Department should
- ✓ Create Successfully

- ✓ Update Successfully
- ✓ Archive Successfully
- ✓ Search Successfully

Module 3 – Employee Management

Module Overview

Employee Management is the core of the application.

Every employee should have a centralized digital profile containing personal information, professional information, documents, reporting manager, attendance history, payroll information and project assignments.

Employee Lifecycle

Candidate

↓

Interview

↓

Offer

↓

Joined

↓

Probation

↓

Permanent

↓

Promotion

↓

Transfer

↓

Resignation

↓

Exit

Employee Information

Personal Details

- First Name
 - Last Name
 - Email
 - Mobile
 - Address
 - Gender
 - Blood Group
 - DOB
-

Professional Details

- Employee ID
 - Department
 - Designation
 - Joining Date
 - Reporting Manager
 - Employment Type
 - Salary Grade
-

Documents

- Aadhaar
 - PAN
 - Resume
 - Offer Letter
 - Certificates
 - Photograph
-

Features

Feature	Description
Add Employee	
Edit Employee	
Archive Employee	
Search Employee	
Upload Documents	
View Timeline	

Feature	Description
Assign Manager	
Generate Employee ID	
Export Employees	

Employee Creation Workflow

```
graph TD; A[HR Login] --> B[Add Employee]; B --> C[Enter Details]; C --> D[Upload Documents]; D --> E[Assign Department]; E --> F[Assign Manager]; F --> G[Generate Employee ID]; G --> H[Create Login]; H --> I[Send Welcome Email];
```

Employee JSON

```
{
  "employeeId": "EMP1023",
  "name": "Rahul Sharma",
  "department": "Engineering",
  "designation": "Software Engineer",
  "joiningDate": "2026-01-15",
  "manager": "EMP005",
  "status": "Active"
}
```

Business Rules

Rule	Description
Employee ID Auto Generated	
Email Unique	
Joining Date cannot exceed current date	
Deleted employees archived	
Salary visible only to HR & Finance	

Validation Rules

Field	Validation
Name	Required
Email	Unique
Salary	Positive Number
Joining Date	Required
Mobile	Valid Number

Notifications

Event	Recipient
Employee Created	Employee
Department Changed	Manager
Manager Changed	Employee
Salary Updated	Finance

Employee Dashboard

Search Employee			
[Rahul] 🔍			

ID	Name	Department	Status
EMP001	Rahul	Engineering	Active
EMP002	Amit	HR	Active
EMP003	Neha	Finance	Active

Acceptance Criteria

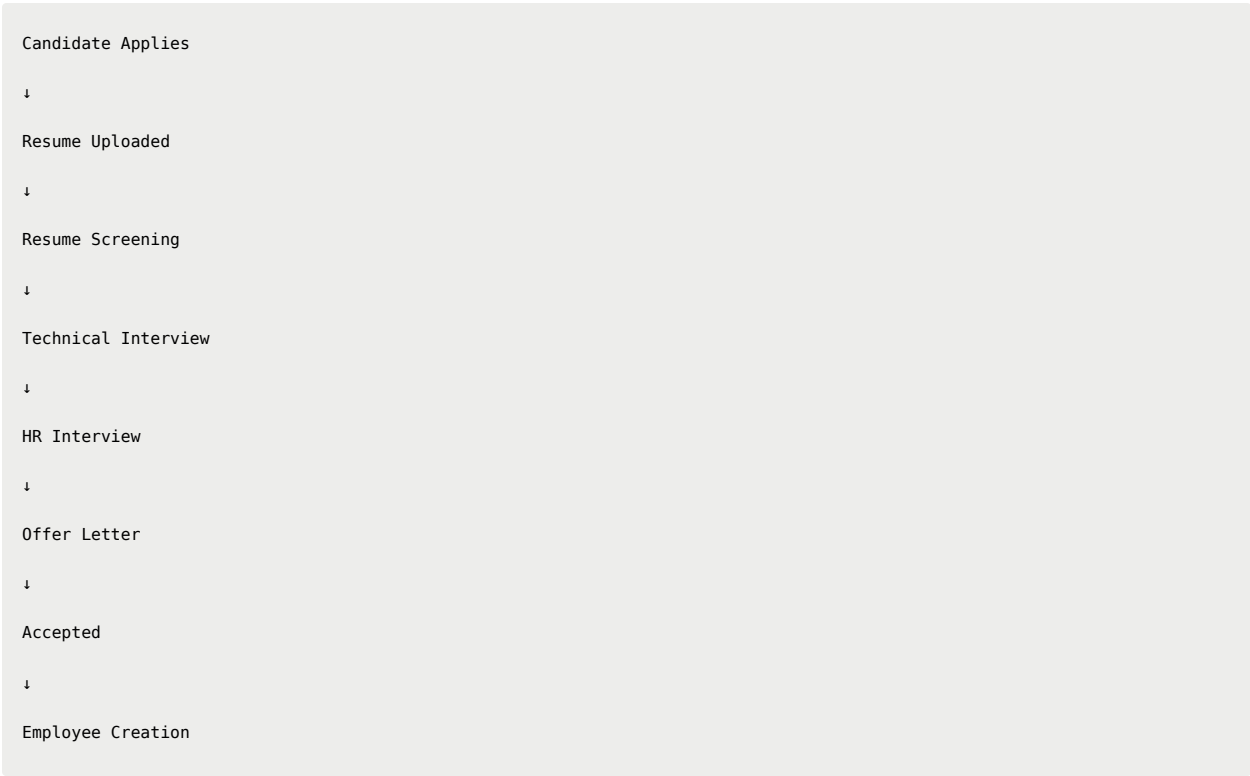
- ✓ Employee Created
- ✓ Documents Uploaded
- ✓ Login Credentials Generated
- ✓ Manager Assigned
- ✓ Audit Log Generated

Module 4 – Recruitment Management

Module Overview

This module manages the complete hiring lifecycle from candidate application to onboarding.

Recruitment Workflow



Features

Feature	Description
Candidate Registration	

Feature	Description
Resume Upload	
Resume Parsing	
Interview Scheduling	
Interview Feedback	
Offer Letter	
Hiring Dashboard	

AI Resume Analysis

The AI assistant should automatically analyze uploaded resumes.

Example

Input

Resume.pdf

Output

Candidate Score : 88%

Skills

✓ React

✓ Node.js

✓ MongoDB

Missing

Docker

AWS

Redis

Candidate JSON

```
{
  "candidateName": "Priya Singh",
  "email": "priya@gmail.com",
  "experience": 2,
  "skills": [
    "React",
    "NodeJS",
    "MongoDB"
  ],
}
```

```
"status": "Technical Interview"
}
```

Business Rules

Rule	Description
Candidate Email Unique	
Resume Mandatory	
Interview cannot be scheduled without screening	
Offer only after HR Approval	

Notifications

Event	Recipient
Interview Scheduled	Candidate
Interview Reminder	Interviewer
Offer Generated	Candidate
Candidate Selected	HR

Recruitment Dashboard

Candidates

Applied : 105

Interview : 42

Offer : 15

Joined : 9

Rejected : 33

Acceptance Criteria

- ✓ Resume Uploaded
- ✓ AI Analysis Generated
- ✓ Interview Scheduled
- ✓ Offer Generated
- ✓ Candidate Converted into Employee

Part 3 – Attendance, Leave, Payroll, Performance & Project Management

Module 5 – Attendance Management

5.1 Module Overview

The Attendance Management module enables employees to record their daily working hours while allowing managers and HR to monitor attendance, working hours, overtime, late arrivals, early departures, and attendance corrections. The system should automate attendance tracking and integrate it with payroll processing.

Business Objective

To eliminate manual attendance tracking, improve attendance accuracy, and provide real-time visibility into employee working hours.

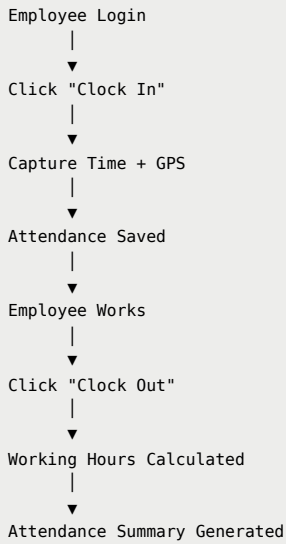
Actors

Actor	Responsibility
Employee	Mark attendance, request correction
Team Lead	Monitor team attendance
Manager	Approve attendance corrections
HR	Manage attendance policies
Finance	View attendance for payroll

Features

Feature	Description
Daily Clock In	Employees record start time
Daily Clock Out	Employees record end time
GPS Verification	Capture employee location during attendance
QR Code Attendance	Office QR-based attendance
Attendance Correction	Request correction for missed attendance
Overtime Calculation	Automatic overtime tracking
Monthly Attendance Report	HR reports
Late Arrival Detection	Identify employees arriving late
Early Exit Detection	Identify early departures

Attendance Workflow



Attendance Status

Status	Description
Present	Employee completed working hours
Absent	No attendance recorded
Late	Clock In after office timing
Half Day	Worked less than required hours
Leave	Approved leave
Holiday	Organization holiday
Work From Home	Remote attendance

Business Rules

Rule ID	Description
ATT-01	Employee can mark attendance only once per day
ATT-02	Attendance must be within office timing
ATT-03	GPS verification required (if enabled)
ATT-04	Attendance corrections require manager approval
ATT-05	Overtime begins after configured office hours

Sample Attendance JSON

```
{
  "employeeId": "EMP001",
  "date": "2026-07-20",
  "clockIn": "09:02",
```

```
"clockOut": "18:15",
"workingHours": 9.13,
"status": "Present",
"overtime": 1.15
}
```

Notifications

Event	Recipient
Attendance Marked	Employee
Late Arrival	Manager
Correction Requested	Manager
Correction Approved	Employee

Acceptance Criteria

- ✓ Employee can mark attendance
- ✓ Attendance automatically calculates working hours
- ✓ Monthly reports generated
- ✓ Overtime calculated
- ✓ Attendance visible in payroll

Module 6 – Leave Management

Module Overview

The Leave Management module enables employees to apply for different types of leave while managers and HR review and approve requests according to company policies.

Business Objective

Digitize leave requests and reduce approval delays.

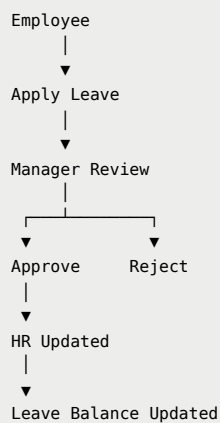
Leave Types

Leave Type	Maximum
Casual Leave	12 Days
Sick Leave	10 Days
Earned Leave	15 Days
Maternity Leave	Organization Policy
Paternity Leave	Organization Policy
Work From Home	Based on Approval

Features

- Apply Leave
- Cancel Leave
- Leave Balance
- Leave History
- Leave Approval
- Holiday Calendar
- Leave Reports

Leave Approval Workflow



Leave JSON

```
{
  "employeeId": "EMP001",
  "leaveType": "Casual Leave",
  "startDate": "2026-08-12",
  "endDate": "2026-08-14",
  "reason": "Family Function",
  "status": "Pending"
}
```

Business Rules

Rule	Description
Leave balance cannot be negative	
Past leave cannot be applied	

Rule	Description
Manager approval mandatory	
HR notified after approval	
Leave deducted automatically	

Notifications

Event	Recipient
Leave Applied	Manager
Leave Approved	Employee
Leave Rejected	Employee
Leave Cancelled	HR

Acceptance Criteria

- ✓ Leave request submitted
- ✓ Leave approved/rejected
- ✓ Leave balance updated
- ✓ Notification sent

Module 7 – Payroll Management

Module Overview

Payroll Management automates salary calculation based on attendance, leave, overtime, bonuses, deductions, and tax components.

Business Objective

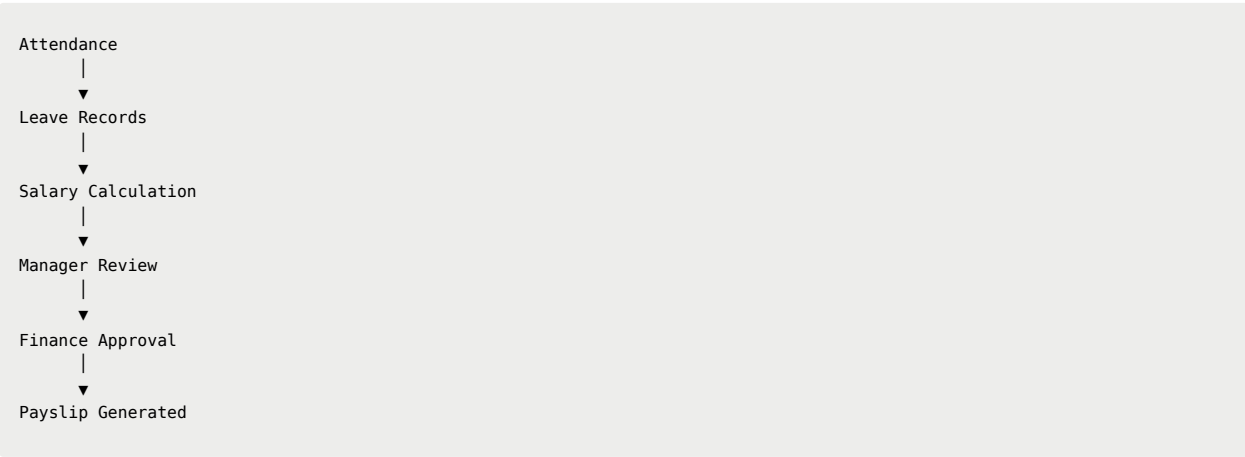
Generate accurate monthly salaries with minimal manual intervention.

Salary Components

Component	Description
Basic Salary	Fixed salary
HRA	House Rent Allowance
Bonus	Performance bonus
Overtime	Extra hours payment
PF	Provident Fund
Professional Tax	State tax
Income Tax	Applicable deductions

Component	Description
Other Deductions	Loans, penalties

Payroll Workflow



Payroll JSON

```
{
  "employeeId": "EMP001",
  "basicSalary": 60000,
  "hra": 12000,
  "bonus": 5000,
  "overtime": 3000,
  "deductions": 2500,
  "netSalary": 77500
}
```

Features

- Monthly Payroll
- Salary Slips
- Bonus Calculation
- Overtime Calculation
- Salary History
- Tax Summary
- Payroll Reports

Business Rules

Rule	Description
Payroll generated once per month	
Salary visible only to HR & Finance	
Attendance mandatory before payroll	
Leave deductions automatic	

Acceptance Criteria

- ✓ Salary calculated
- ✓ Payslip generated
- ✓ Salary history maintained
- ✓ Payroll reports available

Module 8 – Performance Management

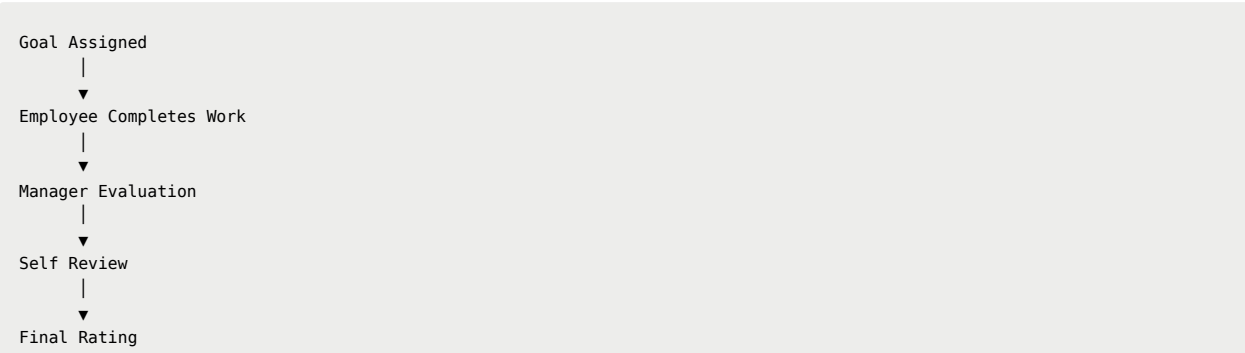
Module Overview

The Performance Management module enables managers to evaluate employees based on goals, KPIs, project completion, attendance, and peer feedback.

Features

Feature	Description
Goal Setting	
KPI Tracking	
Quarterly Review	
Manager Feedback	
Self Assessment	
Promotion Recommendation	
Performance Dashboard	

Performance Review Workflow



↓
Promotion Recommendation

Performance Levels

Rating	Description
5	Outstanding
4	Excellent
3	Good
2	Needs Improvement
1	Unsatisfactory

Sample Performance JSON

```
{
  "employee": "EMP001",
  "quarter": "Q2",
  "kpiScore": 88,
  "attendance": 96,
  "managerRating": 4,
  "overall": "Excellent"
}
```

Acceptance Criteria

- ✓ Goals assigned
- ✓ Reviews completed
- ✓ Ratings generated
- ✓ Dashboard updated

Module 9 – Project & Task Management

Module Overview

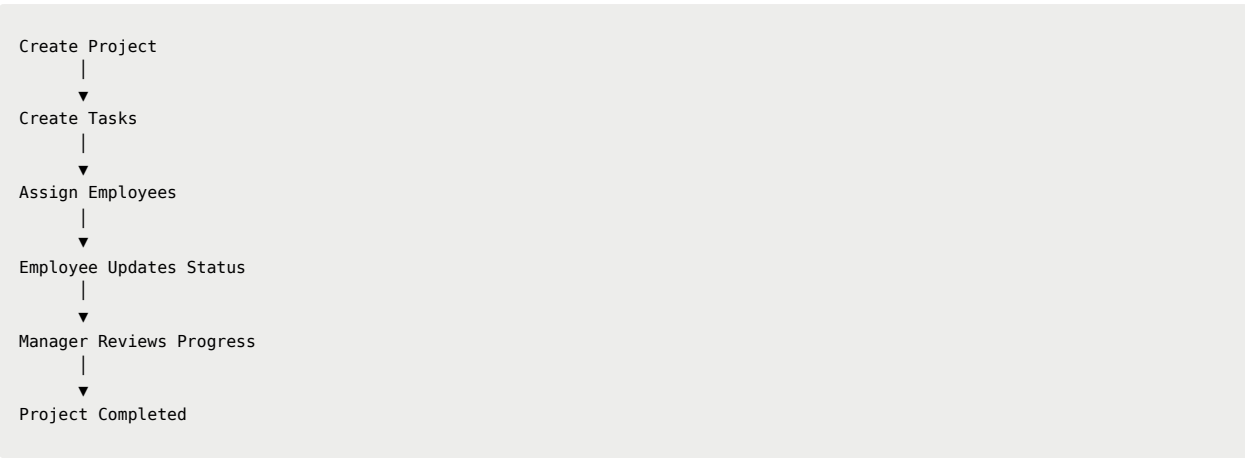
Managers can create projects, assign tasks, monitor progress, and evaluate employee productivity.

Features

Feature	Description
Create Projects	
Create Tasks	

Feature	Description
Assign Team Members	
Task Comments	
File Attachments	
Sprint Tracking	
Kanban Board	
Time Tracking	
Project Dashboard	

Project Workflow



Task Status

Status	Description
To Do	Not Started
In Progress	Currently Working
Review	Awaiting Review
Completed	Finished
Blocked	Waiting on Dependency

Sample Task JSON

```
{
  "project": "Employee Portal",
  "task": "Build Attendance Module",
  "assignedTo": "EMP001",
  "priority": "High",
  "status": "In Progress",
  "deadline": "2026-08-15"
}
```

Business Rules

Rule	Description
One owner per task	
Deadline mandatory	
Task cannot be completed without review	
Completed tasks become read-only	

Notifications

Event	Recipient
Task Assigned	Employee
Deadline Reminder	Employee
Task Completed	Manager
Project Completed	HR

Acceptance Criteria

- ✓ Project created
 - ✓ Tasks assigned
 - ✓ Progress tracked
 - ✓ Dashboard updated
-